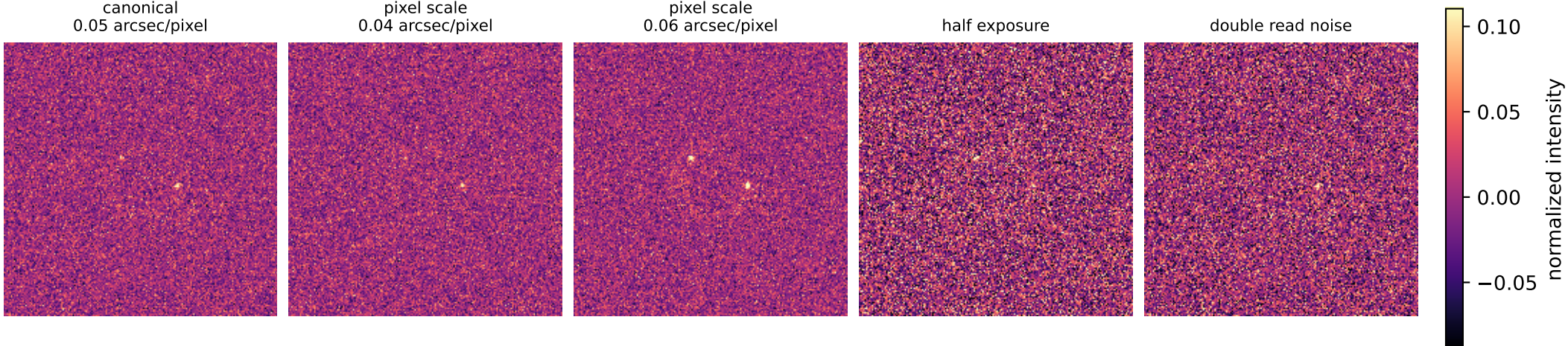
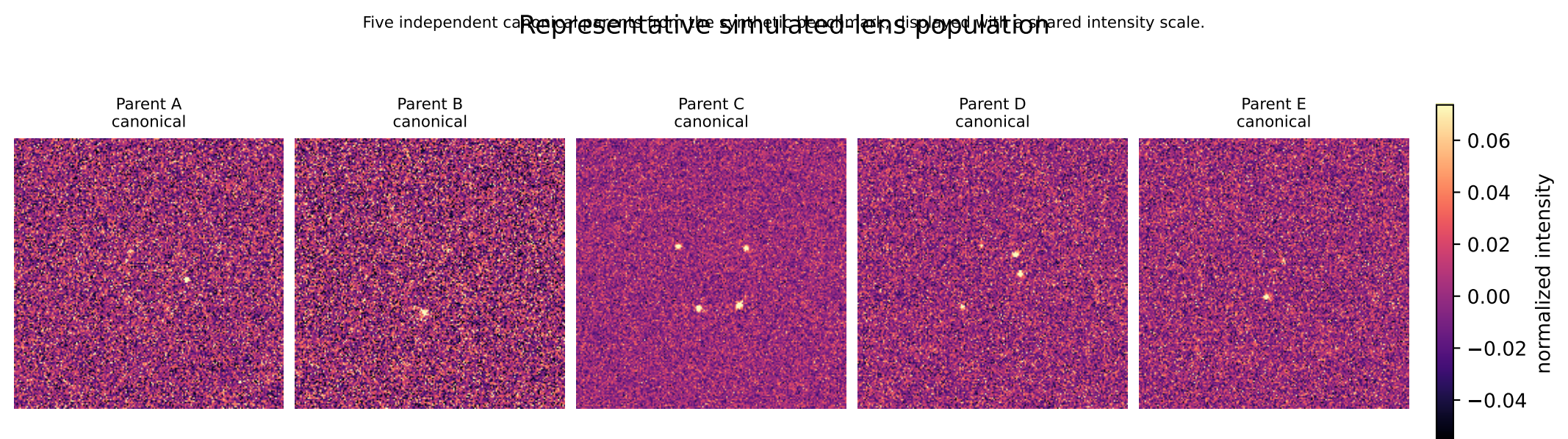
The latent lens/source parameters are held fixed; only the observation condition changes.

One simulated lens under five observation conditions

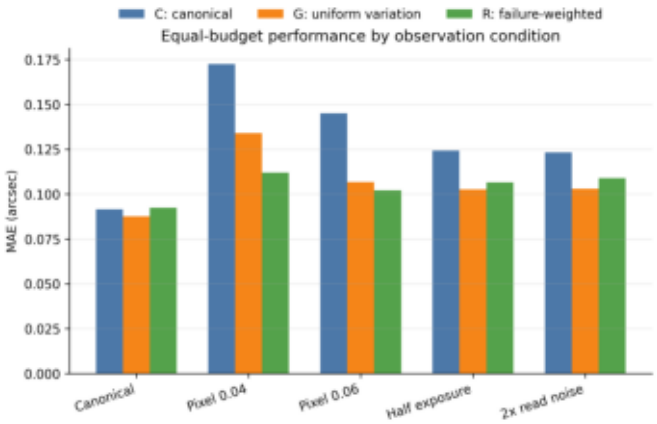


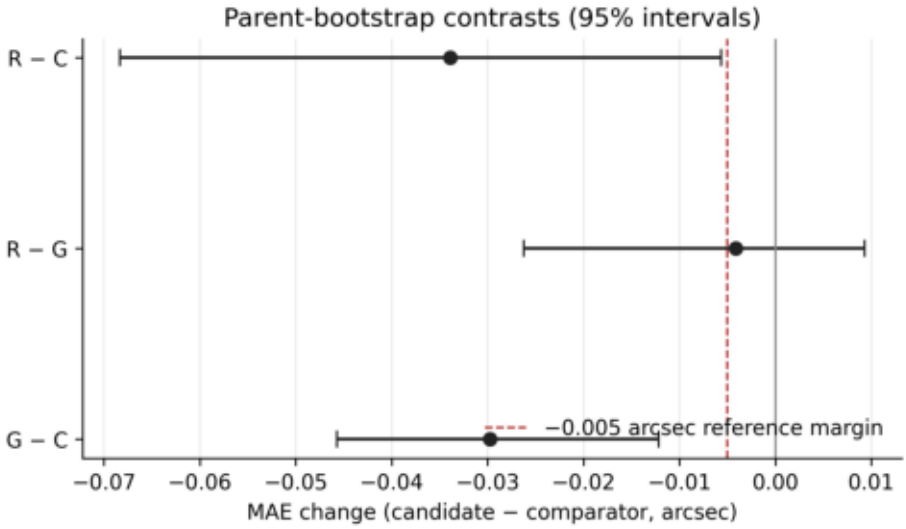

Representative simulated lens population
Five independent canonical parents from the synthetic catalog, displayed with a shared intensity scale.
Parent A
canonical
Parent B
canonical
Parent C
canonical
Parent D
canonical
Parent E
canonical
0.06
0.04
0.02
0.00
−0.02
−0.04
normalized intensity

Equal-budget synthetic robustness results

Frozen parent-disjoint holdout; 150 parents; 10 paired seeds; 880 updates per fit

Arm	Canonical MAE (arcsec)	Four-stress MAE (arcsec)
canonical	0.091536	0.141251
uniform variation	0.087591	0.111507
difficult-condition sampling	0.092366	0.107385





Primary G-minus-C contrast: -0.029744 arcsec; parent-bootstrap 95% interval [-0.045725, -0.012141]. R-minus-G is inconclusive. These are synthetic-only results.

What the study shows and does not show

Synthetic scene: SIE lens + extended Sersic source and lens light + unresolved lensed point source, Gaussian PSF, and detector/noise model.

Intervention: render the same latent parent under canonical, pixel-scale, exposure, and read-noise conditions. Train canonical (C), uniform variation (G), and a source-development policy that samples difficult conditions more often (R) arms with identical CNN architecture, initialization, optimizer, update count, and source exposure.

Leakage control: 697 qualified source parents and a frozen 150-parent holdout; predictions are sealed before target labels are joined; bootstrap resampling is over parent systems, not rendered views.

Interpretation: uniform variation improves robustness on the declared synthetic stress matrix. The difficult-condition policy does not show a reliable extra gain. The AutoLens/Lenstronomy run compares two image-generation implementations; it is not an independent-simulator test. No real-data accuracy claim is made.

Study identifiers: reliability-blind-holdout-v1 / equal-budget-final-v1
This PDF is a visual explanatory supplement, not a new scoring run.